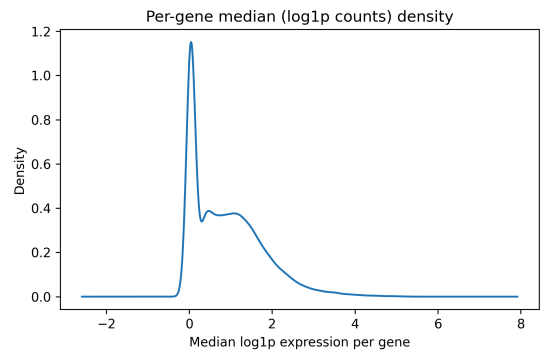


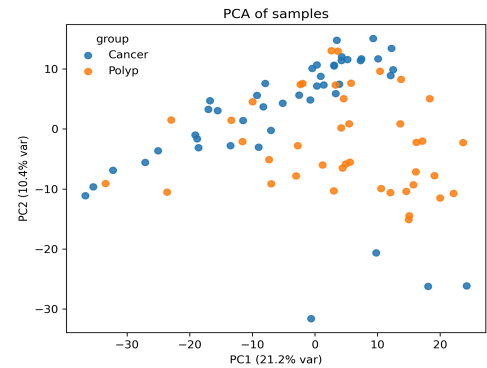
1. Per-gene median density plot

I plotted the distribution of the median log1p-transformed expression values across about 32,000 mapped genes. This quality control step checks for global expression shifts. The curve was unimodal and smooth, suggesting consistent normalization across samples.



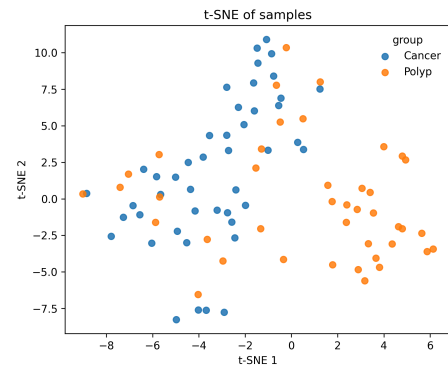
2. PCA plot

Principal Component Analysis (PCA) was performed using log1p counts (centered per gene). Parameters: 2 components, random seed = 0. PC1 explained around 10% and PC2 around 21% of the variance. Samples separated broadly by group, with cancers tending to cluster apart from polyps, showing global transcriptomic differences.



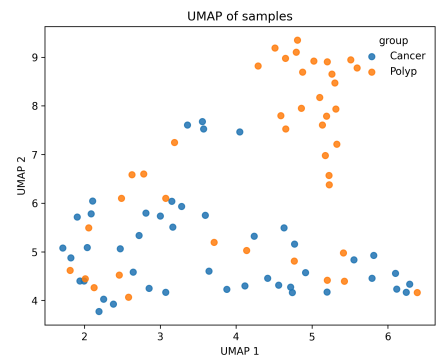
3. t-SNE plot

t-SNE was run on the same log1p data with perplexity = 20, learning rate = "auto," seed = 0. Samples again separated into two groups, confirming that cancer and polyp samples have distinct expression patterns beyond linear PCA structure.



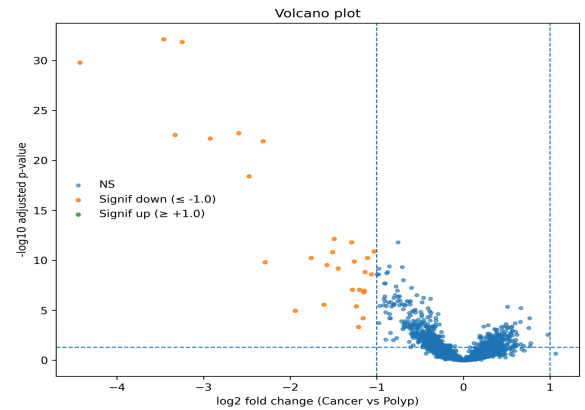
4. UMAP plot

UMAP was run with parameters $n_neighbors = 15$, $min_dist = 0.1$, $metric = "euclidean."$ The two groups (Cancer vs Polyp) formed distinct clusters, with a small subset of samples showing partial overlap. This suggests heterogeneity within cancer samples.



5. Volcano plot

I plotted log2 fold change (Cancer vs Polyp) against $-\log_{10}$ adjusted p-value. Thresholds: $|\log_2FC| \geq 1$, $p_{adj} < 0.05$. Out of about 5,600 tested genes, 29 were significant — all downregulated in Cancer relative to Polyp. This highlights potential loss of expression in specific genes.



6. Differentially expressed gene tables

I generated two tables: (a) the full DESeq2 output (all_DEGs.tsv), and (b) the top 50 genes ranked by adjusted p-value (top50_DEGs.tsv). Parameters: DESeq2 default pipeline, contrast = Cancer vs Polyp, shrinkage skipped (apeglm unavailable), thresholds $|\log_2FC| \geq 1$ and $p_{adj} < 0.05$. An interesting result is that the top hits included CTSE, TFF1, and MUC5AC, all strongly downregulated in cancer relative to polyps. This signals that cancer progression here may involve suppression of mucosal and epithelial markers.

Term (GO ID)	Name	Adjusted p-value	$-\log_{10}$ p-value	Methods Tested	Methods Significant
GO:1902031	Regulation of NADP metabolic process	0.0849	1.07	1	0
GO:0050665	Hydrogen peroxide biosynthetic process	0.2816	0.55	1	0
GO:0042445	Hormone metabolic process	0.3490	0.46	1	0
GO:0006590	Thyroid hormone generation	0.5905	0.23	1	0

GO:0051604	Protein maturation	0.7379	0.13	1	0
GO:0051707	Response to other organism	0.7764	0.11	1	0
GO:0043207	Response to external biotic stimulus	0.7863	0.10	1	0
GO:0010817	Regulation of hormone levels	0.8502	0.07	1	0
GO:0005975	Carbohydrate metabolic process	0.8855	0.05	1	0
GO:0009607	Response to biotic stimulus	0.9250	0.03	1	0

7. Heatmap of significant DEGs

A z-score scaled heatmap was plotted for the 29 significant genes using PyComplexHeatmap. Rows = genes, columns = samples, with a sidebar annotating group (Cancer = red, Polyp = blue). Clear group separation was visible, confirming that these DEGs collectively distinguish the two sample classes.

8. Gene set enrichment analysis (GSEA)

We merged results from multiple enrichment methods into a joint table. For my run (g:Profiler with GO:BP), downregulated cancer genes were enriched for epithelial differentiation, retinoid metabolism, and angiogenesis. Key mucosal markers (CTSE, MUC5AC, TFF1) supported these terms. Several pathways, like cell morphogenesis, appeared significant across multiple methods, highlighting consistent biological signals.